

PANJABUNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (Medical lab Technology)

As per NEP 2020

w.e.f. session 2025-26

Course Outline for B.Voc Medical Lab Technology
as per NEP 2020 w.e.f. session 2025-26

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLT 101	Core Discipline	Basics of Human Anatomy	3	1	4	40	10	40	10	100
MLT 102	Core Discipline	Fundamentals of Microbiology	3	1	4	40	10	40	10	100
MLT 103	Core Discipline	Introduction to Hematology	3	1	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket			2					50
SEC I	Skill Enhancement Course	Common Basket	2	1	3	40	10	20	05	75
MDSC 101	Multidisciplinary Course	Common Basket			3	60	15			75
VAC I	Value-Added Course	Common Basket	2		2	40	10			50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLT 201	Core Discipline	Basics of Physiology	3	1	4	40	10	40	10	100
MLT 202	Core Discipline	Basics of Biochemistry	3	1	4	40	10	40	10	100
MLT 203	Core Discipline	Analytical Laboratory Testing Process-I	3	1	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket			2	40	10			50
SEC II	Skill Enhancement Course	Common Basket			3	60	15			75
MDSC 201	Multidisciplinary Course	Common Basket			3	60	15			75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Medical Lab Technician

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3weeks of the start of 3rdsemester.

** Minimum Credits required to get Certificate Course in Medical lab technology= 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLT 301	Core Discipline	Introduction to Parasitology and Medical Entomology	3	1	4	40	10	40	10	100
MLT 302	Core Discipline	Introduction to Laboratory Equipments	3	1	4	40	10	40	10	100
MLT 303	Core Discipline	Bacteriology, Mycology and Virology	3	1	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket			2					50
SEC I	Skill Enhancement Course	Common Basket	2	1	3	40	10	20	05	75
MDSC 301	Multidisciplinary Course	Common Basket			3	60	15			75
VAC I	Value-Added Course	Common Basket			2	40	10			50
PW II	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLT 401	Core Discipline	Clinical Laboratory Management	4		4	40	10	40	10	100
MLT 402	Core Discipline	Introduction to Histopathology	3	1	4	40	10	40	10	100
MLT 403	Core Discipline	Introduction to Cytopathology	3	1	4	40	10	40	10	100
MLT 404	Core Discipline	Microbiology-I	3	1	4					
AEC / MIL	Ability Enhancement Course	Common Basket			2					50
VAC III	Value-Added Course	Common Basket	2		2	40	10	20	05	75
SIT II*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Medical Lab Technician

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within oneweek of start of 5th Semester. Viva-Voce examination to be held within 3 weeks of the start of 5 th semester.

PROGRAMMEOUTCOMES

PO1. Demonstrate technical skills, social behavior and professional awareness for functioning effectively as a laboratory technician.

PO2.Recognize the impact of laboratory tests in a global and environmental context.

PO3Apply knowledge and technical skills associated with medical laboratory technology for delivering quality clinical investigations support.

PO4: Perform routine clinical laboratory procedures within acceptable quality control parameters in hematology, biochemistry, immunohematology and microbiology.

PO5: Demonstrate technical skills, social behavior and professional awareness for functioning effectively as a laboratory technician.

PO6: Operate and maintain laboratory equipments utilizing appropriate quality control and safety procedures.

PO7: Communicate effectively by oral, written and graphical means.

PO8: Function as a leader / team member in diverse professional and industrial research areas.

PO9: Function in an ethical and professional manner without bias against any ethnicity, race, religion, caste or gender.

PO10: Practice professional and ethical responsibilities with high degree of credibility, integrity and social concern.

Semester I

1	Course Title	**Basics of Human Anatomy**
2	**CourseCode**	**MLT101**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	To understand the organization of body cells, tissues, organs, organ systems, and glands in human body.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.Understand the human body and increase scientific knowledge about different anatomical terms and positions of human body.** **CO2. Get the theoretical as well as practical knowledge regarding basic organization of human body so that they can pursue their studies in Medical Lab Technology** **CO3.Learn basic landmarks and position of various organs along with their structure.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**
12	**Course Content**	Credits 3 Theory: 40 Internal Assessment: 10 Total Marks: 50

UNIT I

• Basic unit of body - Cell
• The anatomical organization of body cells, tissues, organs, organ systems, Membranes and glands in human body.
• Introduction to different types of tissues: Epithelial Tissue, Muscular Tissue, Nervous Tissue
Different types of organ systems.
Brief Introduction of different types of body fluids ,secretions and excretions
• Skeletal system: bones, joints and muscles.

UNIT II

Digestive Organs:
• Tongue
• Teeth
• Oral cavity
• Pharynx
• Oesophagus
• Stomach
• Small intestine
• Large intestine
• Liver, Pancreas and Spleen

UNIT III

Respiratory Organs:
• Nasopharynx
• Oropharynx
• Larynx
• Trachea
• Bronchi
• Lungs
• Thoracic cavity
• Pleura and Pleural cavity
Circulation System: Structure of Heart and Brief introduction of main blood vessels.

UNIT IV

Reproductive Organs Male and Female Gonads
Nervous system and Sense organs : Brief Introduction of Central Nervous System and Peripheral Nervous System
Anatomy of Brain , Spinal Cord, Nerves, Eye, Ear, Olfactory Receptors, Gustatory Receptors
Excretory Organs:
• Cortex and medulla of Kidney
• Ureter
• Urinary Bladder
Urethra (male and female)

PRACTICAL

Credits 1
Practical: 40
Internal Assessment: 10
Total Marks: 50

To Study of various organ systems through Charts and Models

13 Suggested Resources for Reading

1. Anatomy & Physiology: Ross and Wilson
2. Anatomy and Physiology: N Murgesh
3. Anatomy and Physiology for nurses: Evelyn Pearce
4. Anatomy and Physiology for nurses : Sears
5. Anatomy and Physiology for nurses : Pearson
6. Human Anatomy: Harie R. Berasari

1	Course Title	**FUNDAMENTALS OF MICROBIOLOGY**
2	**CourseCode**	**MLT 102**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	**To give an overview of various aspects of General microbiology**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to understand the various aspects of General microbiology**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**
12	**Course Content**	

Credits 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

- **Introduction and brief History, development, scope and applications of Microbiology.**
- **Nature of Microbial World : Prokaryotes and eukaryotes, growth pattern in microbes**
- **Morphology and Structure of Microorganisms**
- **Growth and nutrition of microbes**
- **Safety measures in microbiology.**

UNIT II

- **Morphology, General characteristics & fine structure of bacteria, fungi, actinomycete and algae**
- **Classification of bacteria.**
- **Organization of cell wall, cell membrane, flagella and capsules in bacteria.**

- **Morphogenesis in bacteria, formation of spores and cysts.**

UNIT III

- **General characteristics actinomycete and algae**
- **Morphology, General characteristics and classification of pathogenic fungi.**
- **Classification and general properties of Viruses**
- **Morphology, Pathogenicity and laboratory diagnosis of human viruses.**
- **Bacteriophages: Morphology, multiplication, detection and enumeration.**

UNIT IV

- **Methods of Microbiology isolation of pure cultures, theory and practice of sterilization.**
- **Staining of microbes, Theory of Gram staining.**
- **Preparation, uses and standardization of culture media.**
- **Principles of staining methods and preparation of reagents.**
- **Principles and methods of sterilization.**
- **Uses and mode of action antiseptics and disinfectants.**

PRACTICAL

Credits 1
Practical: 40
Internal Assessment: 10
Total Marks: 50

- **Use of microscope in examination of unstained bacteria, fungi, algae, parasites and stained cell preparations including simple staining,**

- **Gram's staining, acid fast staining, capsule staining, spore staining using prokaryotic and eukaryotic cells, hanging drop preparation.**

- **Preparation of culture media, spread plates, pour plates, selective media, differential media**
- **Separation of pure cultures**

13 Suggested Resources for Reading

1. **Mims' Medical Microbiology Richard Goering, Hazel Dockrell , Mark Zuckerman , Ivan M. Roitt , Professor Peter L. Chiodini Publisher Elsevier Health Sciences**
2. **Ananthnaryans and Paniker's Text book of Microbiology edited by CKJ paniker 7th edition,Publisher Orient Longman**
3. **Roitt's Essential immunology Delves, Peter J.,Martin, seamus J.Burton,Dennis R.Roitt, Ivan M Ananthanarayan and Paniker's Textbookof microbiology Kapil , arti ed**
4. **Microbiology: an Introduction,12th edition Gerard J. Tortora Berdell R. Funke and Christine L. Case**
5. **Parasitology Chatterjee K.D.**
6. **Microbiology Pelczar, Michal J and Others**
7. **Medical microbiology Greenwood David and Other**
8. **Ananthanarayan and Panikar's text book of Microbiology Arti kapil**
9. **Immunology Male David and Other**
10. **Mackie and Mc Cartney practical Medical Microbiology Collee J.G and Other**
11. **Bailey and Scott's Diagnostic Microbiology,13th edition Patricia Tille**
12. **Sherris Medical Micrbiology , 6th edition Pottinger, L. Barth Reller and Charles R. Sterling**
13. **Stephenson Calculations for Molecular Biology**
14. **Gerald Karp Cell Molecular Biology**

Medical Informatics Mohan Bansal **Note: Latest edition of text books may be used.**

1	Course Title	**Introduction to Hematology**
2	**CourseCode**	**MLT 103**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical lab Technician**
8	**Course Objectives**	**To gain understanding of blood and component of blood.** **To gain knowledge of hematological Diseases and hematological Investigations.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1 To get understanding of blood and its components as well as hematological techniques and diseases** **CO2. Differentiate various laboratory test findings with their associated clinical conditions.** **CO3. Be able to distinguish the developmental stages of blood cells. It will also cover Bone marrow examination.** **CO4. Identify the various skills necessary to perform blood counts, evaluate blood elements and report within the stated limits of accuracy.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

- Introduction to hematology and laboratory organization Composition and functions of blood and lymph.
- Detailed study of Hemoglobin and its functions of hemoglobin
- Blood groups including Rh. Factor
- Detailed study of Reticulocytes
- Formation of blood. Morphology of normal blood cells and their identifications
- Hemostasis, Mechanism of blood coagulation. Fibrinolysis.

UNIT II

- Various anticoagulants, their uses, mode of action and their merits and demerits. Normal and absolute in hematology. Quality assurance in hematology**.**

UNIT III

- Descriptive study of RBC abnormalities, Disorders related to RBC Normal white cell count & physiological variation Normal white cell count & physiological variation Hematological Diseases: Anemia and various types of anemia, Thalassemia, Polycythemia, hemolytic disease of new born, multiple myeloma, parasitic infections of blood Leukemia: definition and classification (General & FAB).

UNIT IV

- Laboratory Investigation & Bleeding Disorders
- Laboratory preparation for coagulation tests, Routine coagulation tests, prothrombin time , plasma recalcification time ,partial thromboplastin time , activated partial thromboplastin time, thrombin time, Laboratory diagnosis of bleeding disorders .

PRACTICAL

Credits 1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Cleaning of Laboratory glassware in hematology**
2. **Clinical significance, specimen collection, laboratory investigation & preservation of blood for various hematological investigations.**
3. **Preparations of blood smear.**
4. **Haemocytometery, procedures for cell counts-visual as well as electronic**
5. **Total leukocyte count and Differential leukocyte count.**
6. **Determination of total erythrocyte (RBC) count and platelet counts. Errors involved and mean to minimize such errors. .**
7. **Haemoglobinometery, various methods of estimation of Hb, errors involved and standardization of instrument for adaptation for Hb estimation.**
8. **Romanowsky dyes , preparation and staining procedures of blood smears.**
9. **Laboratory tests for assessing bleeding disorders**

13 Suggested Resources for Reading

1. **Atlas of haematology (5/e) G.A. McDonald**
2. **Clinical Haematology Christopher A. Ludlam**
3. **Practical Haematology J.B. Dacie**
4. **Practical Haematology (8/e) S ir John**
5. **Haematology (International edition) Emmanuel C.Besa**
6. **Haematology (Pathophysiological basis for clinical practice (3/e) Stephen M. Robinson**
7. **Haematology for students Practitioners Ramnik Sood**
8. **Hand book of Medical Laboratory Technology (2/e) V.H. Talib**

Note: Latest edition of text books may be used.

Semester 2

1	Course Title	**Basics of Physiology**
2	**CourseCode**	**MLT 201**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory +15 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	**To understand the basics of physiology of different organ system of body.** .
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to the understand basics of physiology of different organ system of body to help students better understanding medical Lab sciences and lab tests.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**
12	**Course Content**	

Credits 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

- Nutrition- Different types of Nutrients and Vitamins
- Integumentary system: Structure of Skin and its functions.
- Digestive system: Physiology of digestion of food and its absorption.

UNIT II

- Lymphatic system: Different types of body fluids and their functions
- Spleen, lymph node and R.E. system
- Excretory System: Urine formation, osmoregulation and counter current mechanism.
- Cardiovascular system: Origin and regulation of heart beat, cardiac cycle, electrocardiogram, cardiac output, blood pressure and micro-circulation.

UNIT III

- Respiratory system- Transport of O2 and CO2, Oxygen dissociation curve of haemoglobin, Bohr effect, chloride(-) shift, Haldane effect
- Muscular System: Ultra structure and physiological basis of skeletal muscle contractions.
- Reproductive System- Brief Introduction to Female Reproductive System and Male Reproductive System
- Fertilization and Gametogenesis.

UNIT IV

- Neural Physiology- Structure of neuron, resting membrane potential, origin and propagation of impulse along the axon, synapse and myoneural junction.
- Endocrine System- Structure, and functions of hormones of thyroid, parathyroid, adrenal, pineal, hypothalamus, pituitary, pancreas, gonads, thymus.
- Hormones of alimentary canal and kidney.

PRACTICAL

Credits 1
Practical: 40
Internal Assessment: 10
Total Marks: 50

Total Marks: 50

- **To Study of body systems mentioned in theory through Charts.**
- **To demonstrate the separate the plasma and serum from given blood sample.**
- **To study of various endocrine glands through permanent slides /Charts.**

13 Suggested Resources for Reading

1. **Textbook of Medical Physiology Guyton and Hall**
2. **Anatomy & Physiology Ross and Wilson**
3. **Anatomy and Physiology N Murgesh**
4. **Anatomy and Physiology for nurses Evelyn Pearce**
5. **Anatomy and Physiology for nurses Sears**
6. **Anatomy and Physiology for nurses Pearson**
7. **Anatomy and Physiology: Understanding the Human Body Clark**
8. **Physiology & Health Education Gandhi & Goel**
9. **Endocrinology Headley**
10. **Human Physiology Andrew Davis**
11. **Manual of Endocrinology and Metabolism Norman Levin**

Note: Latest edition of text books may be used.

1	Course Title	**Basics of Biochemistry**
2	**CourseCode**	**MLT 202**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	**To gain elementary knowledge of Biochemistry.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.Understand the biochemistry of Carbohydrates, Proteins and Fats and their clinical significance.** **CO2. Evaluate problems & results involving Acid-base and pH chemistry, Atomic structure, Solution and Dialysis** **CO3. Evaluate principles of Biochemistry of Enzymes and Nucleic acids.** **CO4. Learn about Radioisotopes and their uses in Biochemistry.** **CO5. Analyze various problems related to various types of solutions and will be able to prepare various types of solutions.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

- Elementary Knowledge of Inorganic Chemistry: Structure of atom, atomic weight, molecular and equivalent weight. Acids, bases and salts. Acid Base Balance, pH, pH indicators pH-measurement (pH meter, pH paper, universal indicator). Normality, Molarity and Molality, Mole, Molar solutions, normal solutions, buffer solutions, percent solutions, saturated solutions, standard solutions
- Radio isotopes and their use in Biochemistry

UNIT II

- Elementary Knowledge of Organic Chemistry: Organic compounds, aliphatic, aromatic, alcohol, ethers, phenols, acids, etc.
- Elementary Knowledge of Physical Chemistry- Osmosis, osmotic pressure, dialysis, surface tension, diffusion, hypotonic, hypertonic and isotonic solutions. Definition and classification of some colloids and crystalloids.

UNIT III

- Elementary Knowledge of Analytical Chemistry: balances- monopan, twopan, toppan, centrifuges, pH meter, colorimeter, spectrophotometer, fluorimeter, flame photometer, ion selective electrodes, urinometer, chromatograph, electrophoresis, densitometer.
- Basic Steps of Analytic Techniques titrimetry ,photometry, Electrochemistry, Immuno - chemistry

UNIT IV

- Introduction and properties of carbohydrates, proteins and fat.
- Introduction and general properties of Nucleic acids and Enzymes
- Elementary knowledge of Minerals, Electrolytes and hormones
- Therapeutic Drug Monitoring
- Metabolism of carbohydrates, proteins and fat.

Practical

Credits 1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Preparation and storage of distilled water**
2. **Preparation of laboratory reagents and standard solutions, storage of chemicals.**
3. **Units of measurements. S.I. Units, measurement of volume, volumetric apparatus.(pipettes,**
4. **flasks, Cylinders)**
5. **Preparation and standardization of volumetric solutions**
6. **Preparation of buffer solution and measurement of their pH**
7. **To prepare the different concentration of solutions**
8. **To prepare of the 1/10 N HCL**
9. **To find out the normality of given solution**

13 Suggested Resources for Reading

1. **A guidebook to Biochemistry Michael Yudkin**
2. **A Manual of Laboratory & Diagnostic Tests (6/ e) Frances Fischbach**
3. **Biochemistry Voet and Voet**
4. **Biochemistry Stryer**
5. **Biochemistry U. Satyanarayan. & U. Chakrapani**
6. **Clinical Biochemistry Richard Luxton**

7. **Clinical Diagnosis & Management by Laboratory method0 (20/e) John Bernard Henary**
8. **Clinical Biochemistry G. Guru**
9. **Handbook of Biochemistry M.A. Siddique**
10. **Textbook of Medical Biochemistry S. Ramkrishnan**
11. **Biochemical Techniques K. Choudhary**
12. **Text book of Medical Biochemistry Chaterjee & Shinde**
13. **Principles of Biochemistry David L. Nelson**
14. **Textbook of Biochemistry and Human Biology G.P. Talwar**
15. **Textbook of Medical Laboratory Technology Godkar and Godkar**
16. **Outline of Biochemistry Conn Stumpf**
17. **Principles of Internal Medicine Isselbacher**
18. **Proteins and Proteomics : Laboratory Manual Richard J. Simpson**
19. **Purifying Proteins for Proteomics: Laboratory Manual Richard J. Simpson**
20. **Enzymes: Biochemistry, Biotechnology & Clinical chemistry, (2001) Palmer Trevor, Publisher: Horwood Pub. Co., England.**
21. **Outlines of Biochemistry: 5th Edition, Erice Conn & Paul Stumpf ; John Wiley and Sons, USA**
22. **Fundamentals of Biochemistry. 3rd Edition (2008), Donald Voet & Judith Voet , John Wiley and Sons, Inc. USA**
23. **Lehninger, Principles of Biochemistry. 5th Edition (2008), David Nelson & Michael Cox, W.H. Freeman and company, NY.**
24. **Biochemistry: 7th Edition, (2012), Jeremy Berg, Lubert Stryer, W.H.Freeman and company, NY**
25. **Biochemical Methods for Agricultural Sciences – Sadasivam and Manikam. Wiley Eastern Limited, 1992..**
26. **Practical Clinical Biochemistry Harold Varley, CBS; 6 edition (1 December 2006)**
27. **An Introduction to Practical Biochemistry (3rd Edition) – David T Plummer. Tata McGraw-Hill Publishing Company Limited, 1992.**
28. **Biometrics Identity by Sameer Nanawati**
29. **Medicinal Chemistry by AshutoshKar**

Note: Latest edition of text books may be used.

1	Course Title	**Analytical Laboratory Testing Process-I**
2	**Course Code**	**MLT 203**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	**To gain broad understanding of chemicals/reagents useful in sample analysis.** **To get the Knowledge of routine hematological tests from urine, stool, semen and sputum tests.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1. Apply systematized problem solving techniques to identify and correct procedural errors, identify instrument malfunctions and seek proper supervisory assistance, and verify the accuracy of laboratory results obtained.** **CO2. Get broad understanding of chemicals/reagents useful in sample analysis.** **CO3. Operate and maintain laboratory equipment, utilizing appropriate quality control and safety procedure** **CO4. Well versed with the basic knowledge and handling of chemical/reagents doing analysis (tests like hematological tests, urine, stool, semen and sputum tests).** **CO5. To offer the assistance to doctors with clinical laboratory tests, performing the vital role in diagnosis of various clinical conditions.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

- Collection and recording of biological specimens, Collection ,Handling and transportation of Blood Sample,
- Correct method of blood sample collection, Collection methods of samples other than blood samples.
- Examinations of sample for serum Urea and Uric acid.
- Determination of blood sugar level of plasma (or serum) (a) Orthotoluidine method, (b) Glucose oxidase method Determination of the serum urea nitrogen (a) Diacetyl monoxime method Determination of serum creatinine : Alkaline picrate method Determination of serum total cholesterol

UNIT II

- Determination of serum bilirubin a.Malloy and Evelyn b.DMSO method
- Determination of TSH Laboratory investigation for megaloblastic anaemia
- Platelet function tests and their interpretation
- Estimation of Essential electrolytes: Sodium, potassium, calcium, chloride and phosphorus etc.

UNIT III

- Erythrocyte sedimentation rate, factors influencing ESR and various procedures for its estimation with their significance. Haematocrit value by macro and micro methods their merits and demerits. Determination of pack cell volume (PCV) Physiological variations in HB, PCV, TLC and platelets
- Determination and calculation of red blood indices MCH,MCH,MCHC
- Determination of absolute Eosinophil count and Reticulocyte count
- Enumeration of formed elements, Automated systems in hematology

UNIT IV

Urine Examination

- Collection, Handling, transportation of Urine
- Examination of Urine.
- Urine analysis, routine examination of urine(physical examination of urine) rapid chemical tests of Urine
- Determination of specific gravity of urine by urinometer and refractormeter
- Microscopic examination of urine
- Clinical significance and Detailed Examination of Urine
- Biochemical Test Profile (Quantitative determination of Urine)
- Amylase, Calcium, Chlorides, Creatinine, Sodium, Potassium, Glucose, Proteins, Urea nitrogen, uric acid bile pigments, ketone bodies, porphobilinogen, faecal occult blood.

Practical

Credits 1
Theory: 40
Internal Assessment: 10
Total Marks: 50

All Biochemical tests mentioned in Theory

13 Suggested Resources for Reading

1. Textbook of Medical Laboratory Technology Godkar and Godkar
2. Research Methodology in Medical Sciences Chandorkar
3. Practical Clinical Biochemistry Harold Varley
4. Medical Laboratory Sciences, Theory & Practical A. Kolhatkar
5. Medical Laboratory Technology – Volume I Kanai Mukherjee
6. Medical Laboratory Technology – Volume II Kanai Mukherjee
7. Medical Laboratory Technology – Volume II Kanai Mukherjee
8. Medical Laboratory Technology Methods & Interpretation (5/e) Ramnik Sood
9. Enzymes: Biochemistry, Biotechnology & Clinical chemistry, (2001) Palmer Trevor, Publisher: Horwood Pub.

Co., England. Outlines of Biochemistry: 5th Edition, Erice Conn & Paul Stumpf ; John Wiley and Sons, USA
10. Fundamentals of Biochemistry. 3rd Edition (2008), Donald Voet & Judith Voet , John Wiley and Sons, Inc. USA
11. Lehninger, Principles of Biochemistry. 5th Edition (2008), David Nelson & Michael Cox, W.H. Freeman and company, NY.
12. Biochemistry: 7th Edition, (2012), Jeremy Berg, Lubert Stryer, W.H.Freeman and company, NY
13. Biochemical Methods for Agricultural Sciences – Sadasivam and Manikam. Wiley Eastern Limited, 1992..
14. Practical Clinical Biochemistry Harold Varley, CBS; 6 edition (1 December 2006)
15. An Introduction to Practical Biochemistry (3rd Edition) – David T Plummer. Tata McGraw-Hill Publishing Company Limited, 1992

Note: Latest edition of text books may be used.

1	**Course Title**	**Introduction to Parasitology and Medical Entomology**
2	**CourseCode**	**301**
3	**Typology of Course**	**Core Discipline :**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a) The exam of this course will be of 3hours duration. b) There will be no objective type questions. c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	**To Understand the role of parasites and vectors in disease transmission.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to understand the role of parasites and vectors in disease transmission and the most appropriate control strategies.**
10	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1) The syllabus of this paper has been divided into FOUR units.** **2) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4) All questions carry equal marks.**
12	**Course Content**	

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

SECTION I

- **General characters and classification of protozoa of medical importance.**
- **Morphology and life cycles of intestinal Protozon-Amoebic-Giardia.**
- **Laboratory diagnosis of intestinal protozonal infection:- Amoebae-Giardia**
- **Morphology and diagnosis of oral and vaginal flagellates -Trichomonas, - E.gingvalia**
- **Morphology and life cycle of Haemoprotozoa - Malarial Parasite - Laboratory diagnosis of Malarial infection**

SECTION II.

- **Morphology and life cycles of Nematodes (Intestinal) -Ascari -Enterobions - Ancylostoma -Strongyloides-**

Laboratory diagnosis of intestinal Nematode infection.

- **Morphology and life cycle of Haemoflagellates. -Leishmania-Trypansomes, Laboratory diagnosis of Leishmania, Trypanosomes.**
- **Morphology and life cycle of tissue and blood nematodes –Filaria –TrichinellDracuncullus**
- **Lab. Diagnosis of tissue & blood nematode infection. Morphology and life cycle of intestinal cestodes -Taenia-Echinococcus -H.nana-D. latum**
- **Laboratory diagnosis of cesstode infection -Hydatid-Cysticercosis**
- **Life cycle, pathogenic, mechanisms and control of parasitic infections viz. amoebiasis, Kala-azar, toxoplasmosis, ascariasis, filarasis, hook worm infections.**

SECTION III

- **Culture techniques for protozon amoeba, Glardia, Leishmania**
- **Culture methods for Helminths Hookworm round worm.**
- **Egg counting techniques.**
- **Preparation of stains and staining procedures of malaria.**
- **Identification of different plasmodium species.**
- **Preparation of media and maintenance of cultures of E. histolytica.-Giardia -Leishmania**
- **Culture methods for helminthes**
- **Putting up Casoni's test and its interpretation.**
- **Examination of hydatid cyst and processing for preparation of antigen for Casoni's Test.**
- **Examination and processing of Cysticercosis cyst.**

SECTION IV

- **Introduction to Entomology Identification of Adultworms- mosquitoes, flies, ticks and fleas**
- **Animal care, handling and uses in parasitology.**
- **Preparation of parasitic antigens, antigens and antisera**
- **Handling and operating of sophisticated equipment**

Laboratory processing, staining and examination of samples

Practical

Credits :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Introduction to operation of laboratory instruments and safety precautions.**
2. **Macroscopic examination of adult worms, cysts, tissues and processing of stool sample for routine examination.**
3. **Saline preparation for protozoan cysts and trophozoites.**
4. **Concentration procedures for helminthic ova and cyst. Examination and identification of ova and cyst of parasites of medical importance.**
5. **Study of malarial parasite.**
6. **Laboratory diagnosis of kalaazar.**
7. **Detection of trypanosomes(the causal agent of sleeping sickness)**
8. **Laboratory diagnosis of microfilaria(Wuchereria bancroftii)**
9. **Quantitative determination of serum (or plasma) igG class antibodies to toxoplasma gondii by ELISA**
10. **Determination of IgM class antibodies to toxoplasma gondii by ELISA**

13 Suggested Resources for Reading

- **Human Parasitilogy: By (author) Burton J. Bogitsh , By (author) Clint E. Carter , By (author) Thomas Oeltmann 4th Revised edition Publication City/Country San Diego, United States Publisher Elsevier Science Publishing Co Inc.**
- **Clinical Parasitology : A Practical Approach 2nd editionElizabeth A. Zeibig Publisher Elsevier - Health Sciences Division**
- **Veterinary Parasitology M. A. Taylor , By (author) R. L. Coop , By (author) R. L. Wall John Wiley and Sons**

Ltd,3rd revised edition

- **Medical Entomology: A Textbook on Public Health and Veterinary Problems Caused by Arthropods 2nd Edition by B.F. Eldridge (Editor), John Edman (Editor)**
- **Medical and Veterinary Entomology, Second Edition 2nd Editionby Gary R. Mullen (Editor), Lance A. Durden (Editor)**
- **Medical Entomology for Students 3rd Editionby Mike Service (Author)**
- **Text book of Medical Lab Technology,Praful B. Godkar and Darshan P Godkar,PublisherBhalani Publisher,Third edition Vol 1-3**

Note: Latest edition of text books may be used.

1	**Course Title**	**INTRODUCTION TO LABORATORY EQUIPMENTS**
2	**CourseCode**	**302**
3	**Typology of Course**	**Core Discipline :**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a) The exam of this course will be of 3hours duration. b) There will be no objective type questions. c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	• To gain broad understanding of care of laboratory glassware, equipment and instrument • To gain broad understanding of setting up, calibrating, operating, cleaning, maintaining, troubleshooting of laboratory equipment used in quantitative or qualitative analysis • To Calibrate and Validate the Clinical Laboratory instruments and glass wares • To understand Microscopy, working principle, maintenance and applications of various types of microscopes
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to understand the working , care and maintenance of laboratory glassware, equipment and instrument**
10	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1) The syllabus of this paper has been divided into FOUR units.** **2) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4) All questions carry equal marks.**
12	**Course Content**	

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Section I

- **Brief understanding of laboratory planning and laboratory key learning outcome operations.**
- **Laboratory safety rules, hazards and precautions**
- **Handling and Disposal of laboratory waste (Chemical and Biological)**
- **Storage and handling of Chemicals.**
- **Glassware- Description, their uses, cleaning, handling, and care**
- **Laboratory Equipment and their Principles & working: Colorimeter, Autoclave, Centrifuge, Spectrophotometer, Weighing Balance, pH meter, distillation apparatus , Incubator, Hot Air Oven, Water Bath , Anaerobic Jar, , burettes & pipettes , neubauer's Chamber .**
- **Setting up, calibrating and validating, operating, cleaning, maintaining, troubleshooting of laboratory equipment used in quantitative or qualitative analysis**

Section II

- **Techniques of Disinfection**
- **Sterilization: General Principle and methods of Sterilization of glassware & other laboratory instruments**
- **Sterilization of rubber goods, laboratory equipment & other instruments**
- **The cleaning and maintenance procedures of the machine**
- **Identify the cause of errors or other problems or defects in equipments**
- **Glassware – Description of Glassware, its use, handling and care**
- **Importance and methods of cleaning of glass apparatus**
- **Calibration of apparatus and glassware**

Section III

- **Compound microscope: Parts & working**
- **Microscopy, working principle, maintenance and applications of various types of microscopes:-**
  - **Dark ground microscope.**
  - **Polarizing microscope.**
  - **Phase contrast microscope.**
  - **Interference microscope.**
  - **U.V. light microscope.**

SECTION IV

- **Microscopic examination of micro-organism, bright field microscopy, dark field microscopy, phase contrast microscopy, electron microscopy.**
- **Electron microscope: working principle, components and allied techniques for electron microscopy, ultra-microtomy Care and handling of various microscopes-Binocular, DGI, Phase-contrast, fluorescence and electron microscopes**
- **Museum techniques**
- **Lab safety and Instrumentation**

Practical

Credits :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

- **Demonstration of the use of autoclave, centrifuge and spectrophotometer**
- **Cleaning and sterilization of glassware.**
- **Study of the parts of compound microscope.**
- **Using pH meter**
- **Demonstration of setting up, calibrating, operating, cleaning, maintaining, troubleshooting of laboratory equipment used in quantitative or qualitative analysis mentioned in theory paper.**
- **To Calibrate and Validate the Clinical Laboratory instruments and glass wares**
- **To understand Microscopy, working principle, maintenance and applications of various types of microscopes.**

13	**Suggested Resources for Reading:** ➢ **At the Bench : A Laboratory Navigator Kathe Barker** ➢ **At the Helm : A Laboratory Navigator Kathe Barker** ➢ **Basic Medical Laboratory techniques Barbara H. Estridge et al** ➢ **Instrumental Analysis Chatwal Anand** ➢ **Laboratory Reference Jane Roskams** ➢ **Medical Dictionary Oxford** ➢ **Medical Informatics Mohan Bansal** **Note: Latest edition of text books may be used.**	

1	**Course Title**	**Bacteriology, Mycology and Virology**
2	**CourseCode**	**303**
3	**Typology of Course**	**Core Discipline :**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a) The exam of this course will be of 3hours duration. b) There will be no objective type questions. c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	➢ **To learn the techniques of collection of samples, their processing and the identifications of the various pathogens, like bacteria, parasites, viruses, using different techniques.** ➢ **To provide vigorous training in the use of standard safety measures while handing highly infected material.** ➢ **To provide basic knowledge of the different diseases caused by various microorganisms is also imparted.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able** ➢ **CO1 To understand the techniques of collection of samples, their processing and the identifications of the various pathogens, like bacteria, parasites, viruses, using different techniques.** ➢ **CO2 To understand the basic knowledge of the different diseases caused by various microorganisms is also imparted.**
10	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1) **The syllabus of this paper has been divided into FOUR units.** 2) **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3) **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.**

4) All questions carry equal marks.

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Section I

IDENTIFICATION OF BACTERIA:

- **Micrococci, Staphylococci, Streptococci, pneumococci, Corynebacteria, Escherichia, Klebsiella,**
- **E-coli, Enterobacter, Proteus- providencia Salmonella, Citrobacter, Yersinia,**
- **Pseudomonas, Vibrio, Haemophilus, Mycoplasma, Ricketssia, Chalmydia.**
- **Introduction to important diseases caused by Streptococcus, Pneumococcus, Neisseria, Corynebacterium, Bacillus ,Clostridium tetani, Clostridium perfringens, enterobacteriaceae (Proteus, Shigella, Salmonella), Vibrio cholerae, Yersinia, Hemophilus, Mycobacterium, The operative pathogenic mechanisms, laboratory diagnosis, prevention and control of these diseases.**
- **Systematic grouping of pathogenic bacteria.**
- **Laboratory identification of infectious agents, Diagnosis of anaerobic infections ,idenifying characteristics of common pathogenic bacteria,Antimicrobial susceptibility test. IMViC, Urease, catalase, geletineliquification, coagulase, oxidase, sugar fermentation, antibiotic sensitivity test.**

Section II

PATHOGENIC AND NEW-PATHOGENIC FUNGI:

- **Introduction to Human mycotic infections vizCryptococcosis, Dermatophytosis, Blastomycosis, Opportunisitc Mycosis; Candidiasis and Aspergillosis.**
- **Candida, Cryptococci, Dermatophytes, Sporotrichoums, Histoplasma, Blastomyces, Coccidioides, Para-coccidiodes, Dematiaceous fungi, Mycetoma, Actinomyces, Nocardia and common laboratory contaminants. Biochemical tests used for identification of bacteria and fungi.**
- **Preparation and standardization of antigens and antisera.**
- **Lab diagnosis of fungal infections Superficial dermatophyte fungal infections, Candidiasis, creptococosis, Pulmonary infections, Mycetoma, other deep mycotic infections, subcutaneous fungal infections subcutaneous fungal infections spozotrichosis, chromoblastomycosis, Eye and Ear fungi infections**

Section III

VIROLOGY:-

- **Chemotherapy of Viral diseases, Oncogenic Viruses, RNA/DNA Viruses,AIDS, Miscellaneous viruses,**
- **Structure of viruses, lysogenic cycle, lytic cycle, smallpox, polio, HIV,Hepatitis B**
- **Morphology, pathogenesis, life cycle, laboratory diagnosis, prevention and control of viral diseases viz. Rabies, Polio, Small pox, Herpes, Measles, Influenza and AIDS**
- **Introduction to use of different laboratory instruments and their safety precautions. Principles of biosafety hoods use of pipettes, syringes and other virus contaminated**
- **Washing, cleaning and sterilization of Media and glassware in Virology.**

Section IV

- **Instruments in the laboratory. Mode of transmission of viral agents. Prevention of viral diseases. Immunity in viral infection**
- **Collections, handling, and storage of samples for viral diagnosis. Demonstration of preservation of viruses, viral antigens, infects biological materials and viruses.**
- **Different staining techniques used in virology.**
- **Use of Embronated eggs in clinical Virology.**
- **Principles of animal cell culture and their use in virology.**

Practical

Credits :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. Demonstration of staining procedures: Preparation of the following stains and demonstration of viral inclusion bodies:
a) Seller's stain for Negri body demonstration.
b) Giemsa Stain for CMV and Herpes viral inclusions.
2. Preparation of reagents for serological tests:
Phosphate buffered saline, Veronal buffered saline, Alsever's solution and Tris buffer.
3. Principles and performance of viral haemagglutination and Haemagglutination inhibition test.
4. Demonstration of Haemadsorption test, IHA, and RPHA tests.
5. Demonstration of complement fixation test.
6. Demonstration of Immunofluorescence test and lmmunoperoxidase test.
7. Demonstration of ELISA for HbsAg detection.

13 Suggested Resources for Reading

- Mims' Medical Microbiology Richard Goering , Hazel Dockrell , Mark Zuckerman , Ivan M. Roitt , Professor Peter L. Chiodini Publisher Elsevier Health Sciences
- Ananthnaryans and Paniker's Text book of Microbiology edited by CKJ paniker 7th edition,Publisher Orient Longman
- Roitt's Essential immunology Delves, Peter J.,Martin, seamus J.Burton,Dennis R.Roitt, Ivan M Ananthanarayan and Paniker's Textbookof microbiology Kapil , arti ed
- **Microbiology: an Introduction,12th edition Gerard J. Tortora Berdell R. Funke and Christine L. Case**
- **Parasitology Chatterjee K.D.**
- **Microbiology Pelczar, Michal J and Others**
- **Medical microbiology Greenwood David and Other**
- **Ananthanarayan and Panikar's text book of Microbiology Arti kapil**
- **Immunology Male David and Other**
- **Mackie and Mc Cartney practical Medical Microbiology Collee J.G and Other**
- **Bailey and Scott's Diagnostic Microbiology,13th edition** Patricia Tille
- Sherris Medical Micrbiology , 6th edition Pottinger, L. Barth Reller and Charles R. Sterling

Note: Latest edition of text books may be used.

1	**Course Title**	**Clinical Laboratory Management**
2	**CourseCode**	**401**
3	**Typology of Course**	**Core Discipline :**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory + 00 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a) The exam of this course will be of 3hours duration. b) There will be no objective type questions. c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	➢ **To Understand the importance and method of Observing and reporting while dealing with patients**
9	**Course Outcomes**	**On successful completion of the course, the students will be able** ➢ **CO1: To Understand the importance and method of Observing and reporting while dealing with patients** ➢ **CO2 :To Understand Guidelines for Collecting documentation** ➢ **CO3:To maintain restful and working environment**
10	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1) The syllabus of this paper has been divided into FOUR units.** **2) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4) All questions carry equal marks.**

12 Course Content

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

Section I

- **Organization of Laboratory**
- **Functional components of clinical laboratories,(cleanliness, precautions to be taken WRT patients ,reports, analysis**
- **Communication between physician ,patients, and the medical laboratory professional**
- **Basic needs of clinical laboratory technician, awareness of soft skills**
- **How to deal with various people.**
- **Principles and processes for providing customer and personal services including needs assessment techniques**
- **Quality service standards, alternative delivery systems, and customer satisfaction evaluation techniques**

Section II

- **Observing and reporting**
- **Quality control of clinical investigations, Automation in clinical biochemistry laboratory, laboratory organization**
- **Management and maintenance of records.**
- **Understand the importance and method of Observing and reporting while dealing with patients during sample and report collection**
- **The relevant legislation, standards, policies, and procedures followed in the Organization**
- **Role and importance of assisting other healthcare providers in delivering care**
- **Supervise and guide other laboratory personnel**
- **Manage people/patient effectively as per the guidelines.**
- **The principles of leadership and guidance.**
- **Importance and method of Observing and reporting while assisting the pathologists and other members of the team**
- **Importance of verbally informing the person in authority**
- **Research procedures and protocol:**
- **The process of generating or using different sets of rules to combine or group things in different ways**

Section III

- **Documentation**
- **Guidelines for documentation , Guidelines for Collecting documentation**
    - **Various types of records in laboratory set up**
- **Uses and importance of records in laboratory set up**
    - **Essential requirement of records**
    - **Understand abbreviations and symbols**
    - **Enter, transcribe, record, store, or maintain information in written or electronic/magnetic form**
- **Usage of LMIS(Laboratory information management system)**

Section IV

- **Professional Behavior in Healthcare Settings**
- **How to maintain restful environment , Business, mission, and objectives of the organization**
    - **General and Specific etiquettes to be observed on duty**
    - **Understand need for compliance of organizational hierarchy and reporting**
    - **legal and ethical issues**
    - **Importance of conservation of resources in laboratories**
- **Effective working relationships with the people ,The importance of planning, prioritizing and organizing work, Efficient use of time, Importance of keeping the work area clean and tidy**

13 Suggested Resources for Reading

Clinical Laboratory Management	**Lynne Shore**
Medical Laboratory Management	**Sangeeta Sharma et al**
Biostatistics : A Foundation for Analysis in Health Sciences	**Wayne W. Daniel**
Business Communication	**Sinha, K. K.**
Business English	**Taylor, Ken.**
English Vocbulary in Use	**Mccarthy, Micheal.**
Business Letters, Emails	**Shirley Taylor**
The Craft of Business Letter Writing	**Monipally, M. Matthew.**
Presentations Laws	**Anne**
Telephoning and Teleconferencing Skills	**Ken Taylor**
Write to the Point : How to Communicate in business with style and purpose	**Salvatore J. Iacone**
Commercial Correspondence and Office Management	**R. S.N. Pillai**
Comdex Computer Course Kit	**Gupta, V.**
Computer Fundamentals	**P.K. Sinha**
Excel 2010 Inside Out	**Dodge, Mark**
Information Technology for Management	**Lucas, Henry C.**
A Foundation Course in Value Education	**R R Gaur, R Sangal, G P Bagaria, 2009**
Fundamentals of Ethics for Scientists & Engineers	**E G Seebauer& Robert L. Berry, 2000**
Human Values, New Age International Publishers	**A.N. Tripathy, 2003**
Environment Biology	**Agarwal, K. C. 2001**
Environment Protection and Laws	**Jadhav, H &Bhosale, V.M. 1995**
Waste Water Treatment	**Rao M. N. &Datta A.K. 1987**
Environment Science (1st Indian Edition)	**Daniel D. Chiras 2010**
Essentials of Environment Science	**Joseph**
Environment Pollution Control Engineering	**Rao, C.S.**
Perspectives in Environmental Studies	**Kaushik, A.**
Practical Communication Skills	**Chrissie Wright**

Note: Latest edition of text books may be used.

1	**Course Title**	**Introduction to Histopathology**
2	**Course Code**	**402**
3	**Typology of Course**	**Core Discipline :**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a) The exam of this course will be of 3hours duration. b) There will be no objective type questions. c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	➢ **Elementary knowledge of specimen collection** ➢ **Elementary knowledge of tissue fixatives** ➢ **Elementary knowledge of tissue processing**
9	**Course Outcomes**	**On successful completion of the course, the students will be able** ➢ **CO1: To understand the specimen collection , tissue fixatives and tissue processing** ➢ **CO2: To understand the tissue processing: Logging of specimen, preparation of tissues, processing of tissues , Frozen section technique , Handling and embedding of small tissue fragments.** o **Understand about section cutting** o **Understand about Staining** ➢ **CO3: To understand the Staining Procedures** ➢ **CO4: To understand the Auto analyzer, Tissue Processor, Microtome** ➢ **CO5: To understand the Decalcification**
10	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1) The syllabus of this paper has been divided into FOUR units.** **2) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4) All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

SECTION I

- **Introduction to histopathology and laboratory organization.**
- **Elementary knowledge of sample collection**
- **Reception, recording and labelling of histology specimens.**

SECTION II

- **Fixation and various tissue fixatives.**
- **Processing of histological tissues for paraffin-embedding.**
- **Embedding and embedding media.**

SECTION III

- **Microtome-various types, their working principle and maintenance.**
- **Microtome knives and knife sharpening.**
- **Practical section cutting, cutting faults and remedies.**
- **Routine staining procedures, mounting and mounting media.**

SECTION IV

- **Dye chemistry, theory and practice of staining.**
- **Solvents mordents, accelerators and accentutators.**
- **Metachromasia and metachromatic dyes.**
- **Haematoxylin stain. Its importance in histology.**
- **Carbohydrates and amyloid – special stains and procedures.**
- **Connective tissues trichrome staining and other special stains for muscle fibres, elastic, reticulin fibres and collagen fibers.**
- **Principles of metal impregnation techniques.**
- **Demonstration and identification of minerals and pigments**
- **Elementary knowledge of Decalcification**

Practical

Credits :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Tissue processing by using tissue processor**
2. **Sharpening of the microtome knife**
3. **Gross examination and fixation of the specimen**
4. **Decalcification of calcified tissue**
5. **Processing of the tissue by manual method**
6. **Section cutting of paraffin wax embedded tissue**
7. **To fix the section on the slide**
8. **Staining of the tissue section by using hematoxylin and eosin staining method**

13 Suggested Resources for Reading

- Robbins Basic Pathology-Vinay Kumar , Abul K. Abbas , Jon C. Aster
- Histopathology Guy Orchard , Edited by Brian Nation Publisher Oxford University Press
- Text book of Histology,2nd edition,pal GP,Paras medical publisher.
- **Histology for Pathologists** by Stacey E Mills MD
- **Sternberg's Diagnostic Surgical Pathology [2 - Volume Set]** by Stacey E Mills MD

- Anatomic Pathology Board Review, 2e by Jay H. Lefkowitch MD
- Clinical Pathology Board Review, 1e 1 Har/Psc Edition
- **by Steven L. Spitalnik MD (Author), Suzanne Arinsburg DO (Author), Jeffrey Jhang MD (Author)**
- **Medical Laboratory Science-Theory and Practice: J Ochei and A. Kolhatkar**
- **A Hand Book Of Medical Laboratory Technology By V.H.Talib**
- **Inderbir Singh's Textbook of Human Histology with Colour Atlas and Practical Guide** by Vasudeva Neelam
- ***Medical Laboratory Technology* : Methods and Interpretations Vol - 1 6th Edition by *RAMNIK SOOD***

Note: Latest edition of text books may be used.

1	**Course Title**	**INTRODUCTION TO CYTOPATHOLOGY**
2	**CourseCode**	**403**
3	**Typology of Course**	**Core Discipline :**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a) The exam of this course will be of 3hours duration. b) There will be no objective type questions. c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	➢ **To collect exfoliative cytology smears, contact smears and perform applications for cytological examination (under supervision) and carry out routine and special training procedure on cytology smears.**
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** ➢ **CO1: To collect exfoliative cytology smears, contact smears and performs applications for cytological examination (under supervision) and carry out routine and special training procedure on cytology smears.** ➢ **CO2: To organize the histopathology laboratory of the above services and provide basic equipment maintenance.**
10	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1) The syllabus of this paper has been divided into FOUR units.** **2) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4) All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

SECTION I

- **Cytoplasmic constituents and their demonstration**
- **Brief introduction of cytology and cytopathology**
- **Elementary knowledge of sample collection and transportation**
- **Exfoliative Cytology-Specimen Preparation**
- **Elementary knowledge of precautions to be taken for gynecological samples**
- **Elementary knowledge of specimen collection transportation and preservation of precautions to be taken for non gynecological samples**

SECTION II

- **Exfoliative Cytology- Staining Techniques**
- **Understanding of Fluid Specimen**
- **Cytological stains, Papanicolaou stain , other and special stains**
- **Staining techniques**

SECTION III

- **Characteristics of benign and malignant cells**
- **Cervical cytology-basis of detection of malignant and premalignant lesions Hermoral assessment with cytologic techniques and sex chromatin and pregnancy tests.**

SECTION IV

- **Fine needle aspiration cytology (FNAC)**
- **Aspiration cytology principles, indications and utility of the technique with special emphasis on role of cytotechnicial in FNAC clinics**
- **Procedure of fine needle aspiration and section cutting.**
- **Preparation for the procedure**
- **Educate the patient about procedure & Calm down the anxious patients**

Practical

Credits :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Cytochemical staining procedure in various haemopioetic disorders**
2. **Techniques available for cytogenetic studies**
3. **Use of Radioisotopes in hematology**
4. **Safety measures for handling Radioisotopes**

13 Suggested Resources for Reading

- Practical Principles of Cytopathology Revised 1st Edition by Richard M. DeMay (Author)
- Diagnostic Cytopathology: Expert Consult: Online and Print, 3e 3rd Edition by Winifred Gray MB BS FRCPath (Author), Gabrijela Kocjan MD MB BS Spec Clin Cyt (Zagreb) FRCPath(London) (Author)
- Diagnostic Cytopathology Essentials1st Edition Authors: Gabrijela Kocjan Winifred Gray Tanya Levine Ika Kardum-Skelin Philippe Vielh: Churchill Livingstone
- Diagnostic Cytopathology,3rd Edition,**Authors:** Winifred Gray Gabrijela Kocjan Churchill Livingstone
- Diagnostic Cytopathology Essentials: Expert Consult: Online and Print Hardcover – 24 Jun 2013 by Kocjan (Author)

Note: Latest edition of text books may be used.

1	**Course Title**	**Microbiology -I**
2	**CourseCode**	**404**
3	**Typology of Course**	**Core Discipline :**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a) The exam of this course will be of 3hours duration. b) There will be no objective type questions. c) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technician**
8	**Course Objectives**	➢ **To learn the techniques of collection of samples, their processing and the identifications of the various pathogens, like bacteria, parasites, viruses, using different techniques.** ➢ **To provide basic knowledge of the different diseases caused by various microorganisms is also imparted, their processing and the identifications of the various pathogens, like bacteria, parasites, viruses, using different techniques.** ➢ **To provide training in the use of standard safety measures while handing highly infected material.** ➢ **To provide basic knowledge of the different diseases caused by various microorganisms is also imparted**
9	**Course Outcomes**	**On successful completion of the course, the students will be able** ➢ **CO1: To understand the techniques of collection of samples, their processing and the identifications of the various pathogens, like bacteria, parasites, viruses, using different techniques.** ➢ **CO2: To understand the basic knowledge of the different diseases caused by various microorganisms is also imparted, their processing and the identifications of the various pathogens, like bacteria, parasites, viruses, using different techniques.** ➢ **CO3: To understand the use of standard safety measures while handing highly infected material.** ➢ **CO4: To understand the basic knowledge of the different diseases caused by various microorganisms is also imparted**
10	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1) The syllabus of this paper has been divided into FOUR units.** **2) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4) All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Section I

- **Physiology of bacteria, anatomy of bacterial cell, growth requirement of bacteria-growth curve, nutrients required. Gram positive & Gram negative Bacteria.**
- **Elementary knowledge of Pyogenic cocci, Gram Negative Bacilli, Gram Positive bacilli, anaerobes , mycobacteria and spirochetes.**
- **Difference between cocci & bacteria,**
- **Normal flora of human body.**

Section II

- **Lab diagnosis of common Bacterial infections viz:- pyogenic infections, Respiratory tract infections, Meningitis, Diphtheria,**
- **Whooping Cough, Gas gangrene, food poisoning, Enteric fever, Acute diarrhea diseases, cholera, Urinary tract infection,**
- **Tuberculosis, Leprosy, Plague, Anthrax, Typhus fever, syphilis, Gonorrhea and other STD's.**

Section III

- **Disease oriented microbiology, culture & sensitivity test, aerobic, anaerobic techniques**
- **Introduction to Fungi and parasitic fungi, specimen collection, Laboratory diagnosis of mycotic infections, Diagnostic mycology**
- **Principles of Antigen-Antibody reactions.**
- **Principles and mode of action of antibiotics and chemotherapeutic agents for bacteria and fungi.**

Section IV

- **Diagnostic Microbiology & Micro Techniques**
- **Role of microbiology laboratory, specimen handling , laboratory records, safety Regulations, Basic procedures of Diagnostic Rapid and automation methods in Diagnostic Microbiology, Culture environments of microbes , Quality control in microbiology, Quick reference of media and biochemical tests**
- **Collection and handling of faecal specimen, Laboratory techniques in parasitological investigation of stool, Processing of specimens other than stool, Lab identification of human parasites**
- **Collection, transportation and processing of clinical samples for microbiology investigations.**

Practical

Credits :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Preparation of Smear**
2. **Bacteriophage and Bacteriocine typing methods**
3. **Lab diagnosis of common Bacterial infections viz:- pyogenic infections, Respiratory tract infections, Meningitis, Diphtheria,**
4. **Whooping Cough, Gas gangrene, food poisoning, Enteric fever, Acute diarrhea diseases, cholera, Urinary tract infection,**
5. **Tuberculosis, Leprosy, Plague, Anthrax, Typhus fever, syphilis, Gonorrhea and other STD's**
6. **Monochrome staining (simple staining),**
7. **Gram's staining**

8. **Study of motility of capsule**
9. **Study of bacterial capsule**
10. **Study of acid fast bacilli**
11. **Isolation of bacteria by streak plate techniques**
12. **To perform qualitative widal test**

13 Suggested Resources for Reading

- Mims' Medical Microbiology Richard Goering , Hazel Dockrell , Mark Zuckerman , Ivan M. Roitt , Professor Peter L. Chiodini Publisher Elsevier Health Sciences
- Roitt's Essential immunology Delves, Peter J.,Martin, seamus J.Burton,Dennis R.Roitt, Ivan M Ananthanarayan and Paniker's Textbookof microbiology Kapil , arti ed
- **Microbiology: an Introduction,12th edition Gerard J. Tortora Berdell R. Funke and Christine L. Case**
- **Parasitology Chatterjee K.d.**
- **Microbiology Pelczar, Michal J and Others**
- **Medical microbiology Greenwood David and Other**
- **Ananthanarayan and Panikar's text book of Microbiology Arti kapil**
- **Immunology Male David and Other**
- **Mackie and Mc Cartney practical Medical Microbiology Collee J.G and Other**
- **Bailey and Scott's Diagnostic Microbiology,13th edition** Patricia Tille
- Sherris Medical Micrbiology , 6th edition Pottinger, L. Barth Reller and Charles R. Sterling

Note: Latest edition of text books may be used.